COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 / Data Structures and Algorithms	**SECTION:**	CPE21S3
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ACTIVITY NO. 1.1 Basic C++ Programming

1. PROCEDURE OUTPUT

ILO A:

Output:

- Create a function that will take two numbers and display the sum.
- Create a function that will return whether variable A is greater than variable B.
- Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

Modified Code

```cpp
#include <iostream>
int sumOfTwoNum(int fNum, int sNum);
void sumOfThreeNum(int fNum, int sNum, int tNum);
void greaterValue(int fNum, int sNum);
bool booleanTrue (bool a, bool b);
int main(){

    std::cout << "The sum of the two numbers: " << sumOfTwoNum(6,7) << std::endl;
    sumOfThreeNum(6,7,8);
    greaterValue (6,7);
    booleanTrue (true, true);

    return 0;
}
int sumOfTwoNum(int fNum, int sNum){

    int sum = fNum + sNum;

    return sum;

}
void sumOfThreeNum(int fNum, int sNum, int tNum){
    std::cout << "Sum: " << fNum + sNum + tNum << std::endl;
}
void greaterValue(int fNum, int sNum){
    if (fNum > sNum){
        std::cout << "The " << fNum << " is the greater value than " << sNum << "." << std::endl;
    }else{
        std::cout << "The " << sNum << " is the greater value than " << fNum << "." << std::endl;
    }
}
bool booleanTrue (bool a, bool b){

    std::cout << "Operation AND: " << (a&&b) << std::endl;
    std::cout << "Operation OR: " << (a||b) << std::endl;
    std::cout << "Operation NOT: " << (a!=b) << std::endl;

    std::cout << "This session is checked successfully. Thank you for using!" << std::endl;
    return true;
}
```

Output of the Modified Code

```
The sum of the two numbers: 13
Sum: 21
The 7 is the greater value than 6.
Operation AND: 1
Operation OR: 1
Operation NOT: 0
This session is checked successfully. Thank you for using!

--------------------------------
Process exited after 0.01208 seconds with return value 0
Press any key to continue . . .
```

ILO B:
Output:

Section:	**Answer:**
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	N/A
Class Declaration and Method Definition Section	`class Triangle{ private: double totalAngle, angleA, angleB, angleC; public: //constructor Triangle(double A, double B, double C); void setAngles(double A, double B, double C); const bool validateTriangle(); };`
Main Function	`int main(){ Triangle set1(23,67,140); if(set1.validateTriangle()){ std::cout << "The shape is a valid triangle. \n"; }else{ std::cout << "The shape is not a valid traingle. \n"; } return 0; }`
Method Definition	`Triangle::Triangle(double A, double B, double C){ angleA = A; angleB = B; angleC = C; totalAngle = A + B + C; } void Triangle::setAngles(double A, double B, double C){ angleA = A; angleB = B; angleC = C; totalAngle = A + B + C; } const bool Triangle::validateTriangle(){ return(totalAngle <= 180); }`

Table 1-1. C++ Structure Code for Answer

First, aside from the usual header file, there is no global declaration in the file. I noticed how different a class from Python and C++ is though it has their own constructor. C++ has both a public and private while Python has encapsulation. The method definition is a bit confusing to explain as well as the main function.

Table 1-2. ILO B output observations and comments.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C #1:

Code:

```
#include <iostream>
int swap (int a, int b);
int main(){

    swap(26, 28);
    return 0;
}
int swap(int a, int b){
    int temp = a;
    a = b;
    b = temp;

    std::cout << "The original variable: " << b << std::endl;
    std::cout << "The temporary variable: " << temp << std::endl;
    std::cout << "The swapped variable: " << a << "\nA: " << b << " B: " << a << std::endl;
}
```

Modified code:

```
The original variable: 26
The temporary variable: 26
The swapped variable: 28
A: 26 B: 28

--------------------------------
Process exited after 0.1967 seconds with return value 0
Press any key to continue . . .
```

ILO C #2:

Code

```cpp
#include <iostream>
double Faren (double Kelvin);
int main(){
    std::cout << "The total Farenheit is " << Faren(286) << std::endl;

    return 0;
}
double Faren (double Kelvin){
//function that converts Kelvin to Farenheit
    double F = (Kelvin - 273.15) * 9 / 5 + 32;

    return F;

}
```

Output of the Code

```
The total Farenheit is 55.13

--------------------------------
Process exited after 0.01159 seconds with return value 0
Press any key to continue . . .
```

ILO C #3:

```cpp
#include <iostream>
double x (double x1, double x2);
double y (double y1, double y2);
double distance ();
int main(){

    std::cout << "The values of x: " << x(4,6) << std::endl;
    std::cout << "The value of y: " << y(4,6) << std::endl;
    std::cout << "The distance between two points: " << distance();

}
double x(double x1, double x2){
    return (x2 - x1) * (x2 - x1);
}
double y (double y1, double y2){
    return (y2 - y1) * (y2 - y1);
}
double distance (){
    double D = x(4,6) + y(4,6);
    return D;
}
```

Output of the Code:

```
The values of x: 4
The value of y: 4
The distance between two points: 8
--------------------------------
Process exited after 0.2638 seconds with return value 0
Press any key to continue . . .
```

ILO C #4:

???

CONCLUSION:

- Provide the following:
- Summary of lessons learned
- Analysis of the procedure
- Analysis of the supplementary activity
- Concluding statement / Feedback: How well did you think you did in this activity? What are your areas for improvement?

It's a bit confusing on how I can explain it but learning functions can make the code cleaner and well-organized. This way when developing it would be lesser time to update and it won't affect anything on the main function.

On the analysis of the procedure, the function declaration requires two input values that we put on each function (int fNUm, int sNum). In the function (the below the main function) I put the formula for the sum of two numbers (int sumOfTwoNum (int fNum, int sNum)) and sum of three numbers (int sumOfThreeNum (int fNum, int sNum, int tNum) also the boolean (bool booleanTrue (bool a, bool b)) if all of the operations are all true then it would return true and the greater value between the fNUm and sNum (void greaterValue (int fNum, int sNum) I also used if/else if its greater than the other. Back to the main function, I call all the functions that were made and also display the sum of two numbers using cout.

On the analysis of supplementary, on ILO C#1, I declared the function (int swap (int a, int b) before the main function. Then, on the function definition (int swap (int a, int b) I declared the int temp (or temporary variable) to a, and to swap it a = b which means b has now the temporary variable or the swapped variable. I also display (using cout and also I won't type it in the main function) what will happen on the swapped variable which includes the temporary and original variable. Once I call it in the main function I will just type the values and return it to 0. Next is ILO C #2, I first put the function (double Faren (double Kelvin) before the main function. On the function definition (double Faren (double Kelvin)). I put the formula for Fahrenheit (double F = (Kelvin - 273.15) * 9 / 5 + 32 then return it to F, then proceed to use cout to display the total Fahrenheit and type the value (which is the Kelvin) and return it to 0. Lastly on the ILO C #3, I first put the functions for distance (double x (double x1, double x2), double y (double y1, double y2)) and the distance function itself (double distance()) before the main function. Then after the on the function definition I put the formulas for x (return (x2- x1) * (x2- x1)) and y (return (y2 - y1) * (y2 - y1)) this way I the answer would be by square as well as the formula for distance (double D = x(values) + y(values). Back to the main function, I first display using cout about the values of x and y and the distance between two points then it would return 0.

My own feedback: I think I did well on the activity except on the ILO C #4 where I need to modify the code for the parameter, area and which triangle was made. I didn't do it since it was confusing on how I can add in the class and also caused errors when running the code. Also I need areas of improvement on how to code the formulas since I've done a lot of trial and errors to see if the code is running with that kind of concept.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT, Brave Leo**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of concepts, idea generation**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used ChatGPT and Brave Leo to help me explain what is the code analysis for a specific code.**

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